Karan Kapur

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Languages: English & Hindi (fluent) | Punjabi (native) | Mandarin Chinese (basic conversational)

EDUCATION

University of Toronto — ***Toronto, Canada - Dec 2023***

Master of Engineering: Emphasis in Analytics/Machine Learning (ML)

• Junior Fellow at Massey College • Program Director at Sustainable Engineers Association (SEA)

Thapar Institute of Engineering and Technology (TIET) — ***Patiala, India - Jun 2021***

Bachelor of Engineering: **Chemical Engineering**

• Research Assistant at Defence Research and Development Organisation (DRDO) • Founding Secretary at Association of Computing Machinery (ACM) Student Chapter TIET

PROFESSIONAL EXPERIENCE

NXP Semiconductors — ***Hyderabad, India – Oct 2026 to Present***

Lead Software Engineer - Edge AI & Optimization

- Focus on quantization and model optimization for CNNs, LLMs, and VLMs, targeting efficient on-device inference on NXP's Ara2 NPUs.
- Develop research prototypes and Python/C++ implementations, evaluating accuracy, latency, and memory trade-offs in collaboration with hardware, compiler, and runtime teams.

Himax Technologies — ***Tainan, Taiwan - Mar 2024 to Sep 2026***

Senior Engineer – AI & Deployment

- Engineer and optimize computer vision, audio, and ML models for ARM-based embedded edge ASIC platforms, spanning signal-processing and learning-based architectures under strict memory, latency, and accelerator-throughput constraints.
- Profile and optimize CPU/NPU ML workloads using Python, PyTorch, and TensorFlow across CNNs, transformers, and lightweight models; apply quantization-aware methods, kernel tuning, latency profiling, and quality–power–performance trade-off analysis for real-time execution.
- Design and implement performance-sensitive embedded systems in C/C++, including a JPEG encoder with adaptive quantization/rate control and a data-over-sound modem transferring 1 MB payloads; architect buffer-aware pipelines, port full algorithm stacks to sub-2 MB SRAM MCUs, and tune DSP for deterministic throughput under memory/I/O constraints.
- Work with Tier-1 industrial partners and cross-functional hardware, firmware, product, and runtime teams to convert system requirements into algorithmic decisions; deliver technical training to 30+ engineers and present product-ready Edge AI demonstrations at global forums.

Southern Taiwan Technology Council (STTC) — ***Tainan, Taiwan - Jan 2026 to Present***

Co-Founder

- Built a community-led technology platform engaging participants from 11 countries across 15+ technical/professional fields.
- Lead the programming and delivery of panels, webinars and networking events spanning AI, semiconductors, sustainability, manufacturing and technology entrepreneurship.

Corporate Knights — ***Toronto, Canada – Aug 2023 to Feb 2024***

AI Consultant (Contract)

- Designed and implemented NLP pipelines utilizing OpenAI's GPT-4 as the foundational model (via LangChain and RAG) to upgrade research methodologies and deliver human-validated insights to a real-time database used by client organizations.
- Employed prompt engineering, fine-tuning, and Multi-source and Multi-step Information Extraction (MIE) achieving 92% data validation success rate.
- Developed strategic initiatives to bridge data gaps and explore AI-driven opportunities within Corporate Knights taxonomy efficacy.

AWARDS

- Jackman and Lupina Bursaries – Massey College, University of Toronto — ***2023***
- MITACS Globalink Research Award Abroad - National Cheng Kung University — ***2023***
- Hudd Non-OSOTF scholarship & Lochhead Scholarship - Massey College, University of Toronto — ***2022***
- Merit III Scholarship Recipient -TIET, Patiala, India — ***2017-2020***